Design of diffractive optical elements for high-power laser applications

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Abstract. Diffractive optical technology is rapidly becoming an indispensable tool for the creation of high-efficiency optical beam-shaping devices within a wide range of industrial and research activities. The high damage thresholds of typical materials used in diffractive optical element fabrication make them highly suited for deployment in high-power laser systems. We present two high-power applications of diffractive optical elements designed using an ultrahigh fidelity, ultrahigh efficiency design technique and demonstrate the advantages, in terms of optical alignment, that can be achieved by combining the diffractive structures with a multilevel diffractive lens. © *2004 Society of Photo-Optical Instrumentation Engineers.* [DOI: 10.1117/1.1803553]

Subject terms: diffraction; high-power lasers; beam shaping.

Paper 0431111 received Dec. 1, 2003; revised manuscript received May 3, 2004; accepted for publication Jun. 30, 2004.

1 Introduction

The use of diffractive optical elements in free-space laser beam shaping is a relatively mature exploitation of the technology. Many industrial and medical applications have benefited from the ability of diffractive optics to produce arbitrarily shaped output beams. The use of these elements within high power (>100 MW) laser systems has enabled improvements to the operational efficiency of these systems by, e.g., decreasing the rest time between shots of a Nd:YAG glass laser due to greater energy extraction through the amplifier crystals. Furthermore, the ability of diffractive optics to produce arbitrary beam shapes has driven the creation of elements to produce more esoteric beam shapes, e.g., a cosine-squared distribution for inertial confined fusion experiments.

The computational methods used to design diffractive optical elements (also known as diffractive phase elements or DPEs) are numerous,[1] ranging from closed-form solutions of the Fraunhofer diffraction integral to gradient descent algorithms based around the fast Fourier transform. Generally, the problem to be optimized is the solution of the diffraction integral under some form of output or phase constraint and the optimality of a particular solution can be assessed by using a number of different metrics.[2] The most important of these metrics is the diffraction efficiency (η), which is the proportion of the incident beam energy that ends up in the useful diffraction orders specified by the initial design. The diffraction efficiency of a DPE that is to produce a pattern containing the diffraction orders $\mathcal{M}$ is given by

$$\eta = \sum_{i,j \in \mathcal{M}} U_{ij} U_{ij}^{*}, \tag{1}$$

where U_{ij} is the normalized amplitude of the ijth diffraction order, and $\Sigma_{i,j \in \mathbb{R}^2} U_{ij} U_{ij}^{*} = 1$. The range of diffraction efficiencies that can be realized is primarily determined by the number of levels used for the phase quantization of the element and the ratio of the number of diffraction orders in the output pattern to the number of pixels in the DPE. Typically, a binary (i.e., two-phase-level) design will give a diffraction efficiency of the order of 60 to 70%, with a maximum efficiency of 40.5% in a single diffraction order, whereas an eight-phase-level design will give a diffraction efficiency in the 75 to 85% range. The diffraction order-to-pixel-number ratio will usually be in the range 0.2 to 0.3. The other metrics used to characterize the DPE measure the fidelity of the actual output to the desired output. Two methods are used to quantify this fidelity: the reconstruction or nonuniformity error and the mean square error. The reconstruction error is a measure of the maximum variation of the actual output from the desired output and is defined as

$$\Delta_r = \max_{i,j \in \mathcal{M}} \left[1 - \frac{U_{ij} U_{ij}^{*} N(\mathcal{M})}{\eta I_{ij}} \right]^2, \tag{2}$$

where $N(\mathcal{M})$, is the number of orders in the set of useful diffraction orders, and I_{ij} is the target intensity in the ijth diffraction order. The acceptable values of reconstruction error are very dependent on the application in which the DPE will be used. However, generally a reconstruction error of less than 5% is sufficient for most applications. The mean square error (MSE) of the DPE output is given by,

$$\mathrm{MSE} = \sum_{i,j \in \mathcal{M}} \left[\frac{I_{ij} N(\mathcal{M})}{\Sigma_{k,l \in \mathcal{M}} I_{kl}} - \frac{U_{ij} U_{ij}^{*} N(\mathcal{M})}{\eta} \right]^2. \tag{3}$$

As for the reconstruction error, the range of acceptable values for the MSE is dependent on the precise application in which the DPE is to be used. Typically, an average MSE of around 1 to 2% per diffraction order is sufficient for most applications.

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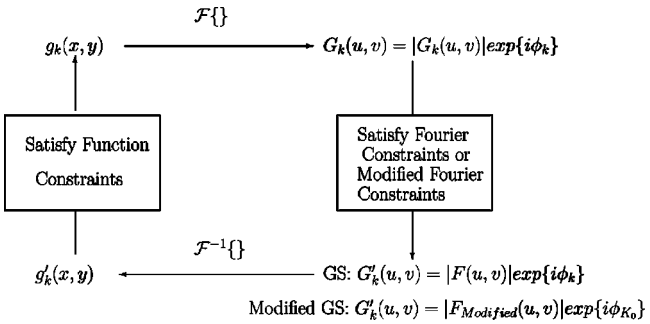


Fig. 1 Comparison of the basic GS algorithm and the modified GS algorithm. The difference is at the second step. In the new algorithm, a modified Fourier-domain constraint function instead of an ideal output function is used.

In this paper, we present a modified iterative Fourier transform algorithm capable of producing high-efficiency, ultra-low-error output beam shapes. In particular, we look at the deployment of these elements within two different high-power laser system applications. The first, which provides beam shaping for a high-power Nd:YLF glass laser, produces a super-Gaussian (SG) flat-top beam. The second, a free-space beam-shaping element, is used to condition and fan-out the output from a 300-mJ solid state laser for launch into multimode optical fibers without damaging the front face of the fibers.

2 Free-Space Laser Beam Shaping

2.1 *An Improved Gerchberg-Saxton Algorithm*

The Gerchberg-Saxton[3] (GS) algorithm with its modifications, e.g., input-output[4] is one of the most effective methods for the design of the DPEs. However, it has serious convergence problems, and, in many cases,[4] is incapable of producing a solution that has a low MSE, especially when the output beams are desired to be diffraction-limited.[5] In this section, we present an improved algorithm that is a modification to the GS algorithm that can lead to a near-optimal design.[6] This improved algorithm can be regarded as a development of the method presented by Prongué et al.[7] The algorithm begins with a phase approximation of the diffractive phase element. This approximation is usually obtained by a reasonable number of the basic GS iterations, i.e., until a stable output beam is achieved. The principle of the modified algorithm is that, since the Fourier-domain constraint, $F(u,v)=[I_{\text{output}}(u,v)]^{1/2}$, fails to satisfy the output constraint, an amplitude-symmetric function of the imperfect output about the ideal Fourier-domain constraint, $|F(u,v)|$, without the change of phase may rectify this. The new iterations are similar to those of input-output[4] but have two different features. First, the phase distribution at every new iteration begins with the results of GS and remains unchanged. Second, while the input-output differs from GS in the object-domain operation, the new iterations differ from GS in the Fourier domain. The modified Fourier-domain constraint function is based on the difference between the output results of GS or the output results of every new iteration, and the required ideal output profile. Figure 1 shows the difference of the new iterations and the basic GS. We can see that the difference is at the second step. In the new algorithm, a modified Fourier-domain constraint function instead of an ideal output function is used.

The new algorithm is employed for the design of a DPE for a square SG beam shaping. The input plane amplitude profile is a Gaussian beam, $G_{\text{input}}(x,y)=\exp\{-[(x^2+y^2)/r_i^2]\}$, where $r_i=2.0$ mm and the element diameter D is chosen to be 10 mm. For wavelength $\lambda=0.532$ μm and focal length $f=75.3$ mm, the focal plane spatial resolution will be $\lambda f/D=4.0$ μm and the far-field spatial extent is $\sim$4.1 mm for a grid of 1024×1024 points. The designed far-field intensity profile is a 200th-power SG beam,

$$I_{\text{output}}=\exp\{-[(x^2/r_f^2)^{200}+(y^2/r_f^2)^{200}]\}, \tag{4}$$

where x_f is set to 250 μm.

Starting with the initial phase distribution obtained by geometrical ray-tracing method, we use basic GS iteration equations

$$G_k(u)=\mathcal{F}[g_k(x)]=\mathcal{F}([I_{\text{input}}(x)]^{1/2}\exp\{i\phi[g'_{k-1}(x)]\}), \tag{5}$$

$$\begin{aligned}G'_k(u)&=|F(u)|\exp\{i\phi[G_k(u)]\}\\&=[I_{\text{output}}(u)]^{1/2}\exp\{i\phi[G_k(u)]\},\end{aligned} \tag{6}$$

$$g'_k(x)=\mathcal{F}^{-1}[G'_k(u)], \tag{7}$$

$$\begin{aligned}g_{k+1}(x)&=|f|\exp\{i\phi[g'_k(x)]\}\\&=[I_{\text{input}}(x)]^{1/2}\exp\{i\phi[g'_k(x)]\},\end{aligned} \tag{8}$$

where k is the k'th iteration, and $\phi(*)$ is the argument of a complex function.

A stable output beam profile can be obtained after K_0 iterations. The phase distribution of the diffractive element obtained from the K_0'th iteration is

$$\phi_{k_0}=\phi[g'_{K_0-1}(x)]. \tag{9}$$

The amplitude distribution of the output beam can be calculated from Eq. (3):

$$|G_{k_0}(u)|=|\mathcal{F}([I_{\text{input}}(x)]^{1/2}\exp\{i\phi[g'_{K_0-1}(x)]\})|. \tag{10}$$

Figure 2 shows the intensity distribution of the output beam $|G_{50}(u)|^2$ after 50 iterations. It is obviously far from a flat-top SG beam. Based on the difference between $|G_{50}(u)|$ and the SG beam $F(u)=[I_{\text{output}}(u)]^{1/2}$, a constraint function is constructed and imposed in the following new iterations. As we mentioned previously, an amplitude-symmetric function of the imperfect output about the ideal Fourier-domain constraint, $|F(u)|$, without the change of phase may produce an improved design. Therefore, a logical choice for the modified Fourier-domain constraint is

$$F_{K_0+k}^{\text{modified}}(u)=\begin{cases}\left|\dfrac{F(u)}{G_{(K_0+k)_N}(u)}\right|^{c_1} & \text{if } u\in\gamma-\delta\\ |F(u)| & \text{if } u\notin\gamma-\delta,\end{cases} \tag{11}$$

or

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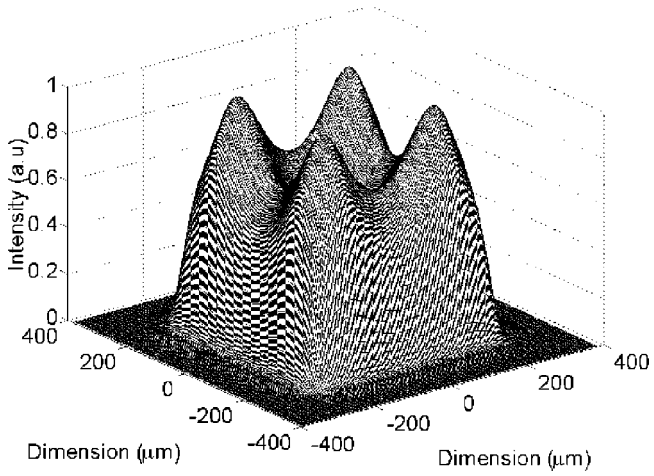


Fig. 2 Far-field intensity distribution of the designed SG beam after 50 GS iterations. The beam top is far from flat and there is an intensity decrease in the beam center.

$$F_{K_0+k}^{\text{modified}}(u)=\begin{cases} c_2|F(u)|-|G_{(K_0+k)_N}(u)| & \text{if } u\in\gamma-\delta \\ |F(u)| & \text{if } u\notin\gamma-\delta. \end{cases} \tag{12}$$

where γ is the set of points at which $G(u)$ violates the Fourier-domain constraints; and δ is a subset of the set γ, where the points of the subset γ are located at the boundary of the region to which the points of the set γ correspond. Both c_1 and c_2 are constants and c_2 is larger than 1. The function of the subset δ is to make the modified $F(u)$ to be only imposed at the center part of the region where the points violate the Fourier-domain constraints. The constraint function is substituted for the SG function $F(u)$ in Eq. (6), the phase remains unchanged and is that of the output of the conventional GS process,

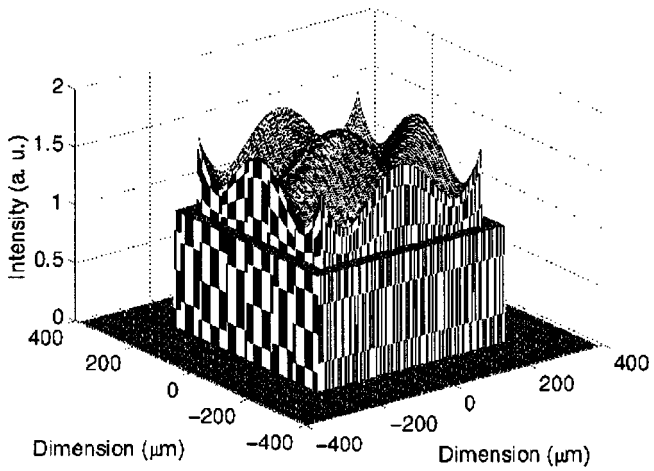


Fig. 3 Modified Fourier-domain constraint function. Basically the intensity distribution of the modified Fourier-domain constraint function is symmetrical with that of Fig. 2 about the ideal SG beam, i.e., the position where a high intensity exists in Fig. 2 corresponds to a lower intensity in the modified Fourier-domain constraint function.

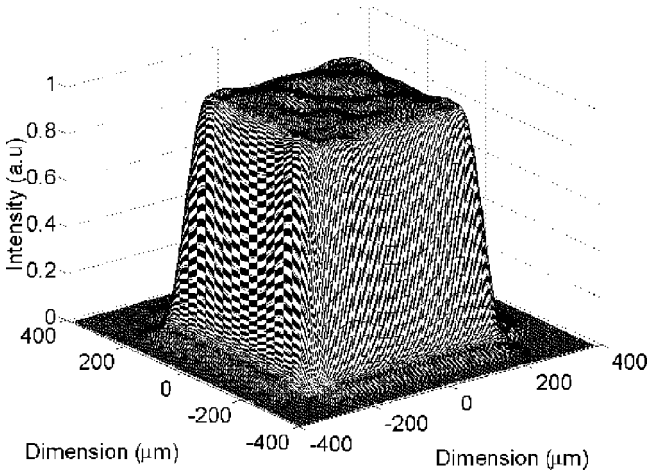


Fig. 4 Final designed far-field output beam profile after using four new iterations. The beam intensity is uniform across the top and decreases to zero abruptly at the edges of the target region. The MSE of the intensity distribution fitted to a 12th-order SG beam 9.75×10^{-3}. The bucket energy contained within the desired region is 97.4%.

$$G'_{K_0+k}(u)=|F_{K_0+k}^{\text{modified}}(u)|\exp\{i\phi[G_{K_0}(u)]\}$$
$$=(I_{K_0+k}^{\text{modified_output}})^{1/2}\exp\{i\phi[G_{K_0}(u)]\}. \tag{13}$$

The $F_{K_0+k}^{\text{modified}}(u)$ can be obtained from either Eq. (11) or (12) and Eq. (5). The new iteration uses Eq. (11) or Eqs. (12), (5), (13), (7), and (8). Essentially, the role of the new iteration process is to look for a new $F_{\text{modified}}(u)$ to obtain a satisfactory output $G(u)$. It is necessary for the phase to remain unchanged, otherwise the energy that leaks outside the desired region will increase. For chosen parameters c_1, c_2, and δ after each iteration, the MSE of the output intensity decreases rapidly at every iteration, and after around four iterations, a satisfactory output has been reached. Figures 3 and 4 show the modulus square of the modified

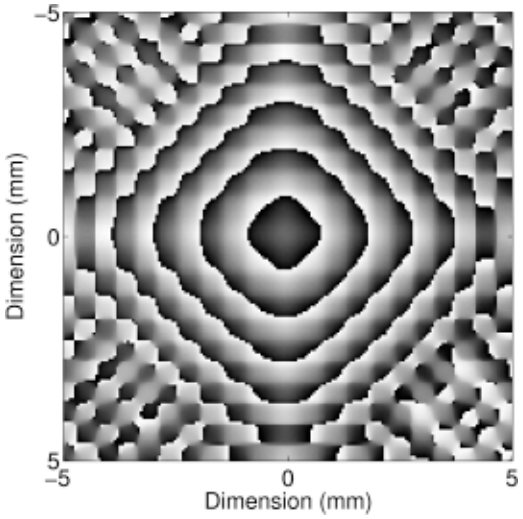


Fig. 5 Far-field beam shaping elements for square output beam in Fig. 4.

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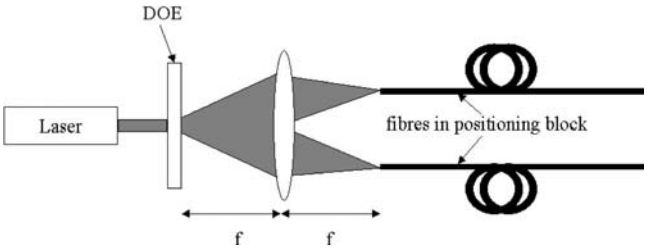


Fig. 6 Schematic of the optical setup comprising a DOE and Fourier-transforming element to produce a compact arrangement.

Fourier-domain constraint function $|F_{\text{modified}}(u)|^2$ and the final output beam profile $|G(u)|^2$ after four new iterations respectively. As we can see, the modified Fourier-domain constraint function and the GS iteration result shown in Fig. 2 are almost symmetrical about the ideal SG beam profile. Figure 4 shows that the intensity distribution remains uniform across the top and decreases abruptly to zero at the edges. In this designed SG beam, about 97.4% of the energy was constrained within the desired region. The MSEs of the amplitude and the intensity of the entire beam are 0.035 and 9.75×10^{-3}, respectively, when fitted to the corresponding parameters of the 12th-power SG beam. Figure 5 shows the far-field element producing the output beam in Fig. 4.

3 Fourier Plane Beam-Shaping Elements for High-Power Fiber Coupling

One of the major limiting factors for fiber delivery of high-peak-power laser beams is the maximum allowable intensity before dielectric breakdown occurs at the input face of the fiber.[8] Using a diffractive beam shaper to create a flat top illumination profile can maximize the throughput of the fiber before this breakdown occurs. In addition to this, it is sometimes desirable to deliver the light through several fibers either to supply multiple processes or to increase the amount of light reaching a single process. Two ways of doing this are to use either a fiber bundle illuminated by a single flat top beam or to combine a flat top beam shaper with a fan-out element to couple to several individually mounted fibers. The use of a fiber bundle leads to a large amount of wasted energy from light that couples into the fiber cladding or the gaps between the fibers.

In this section, we consider the use of Fourier-plane beam-shaping elements[9] to produce the combined fan-out and beam-shaping effect. Fourier beam-shaping elements produce an image consisting of discrete, evenly spaced diffraction orders. These can be made to mimic uniform illumination by packing them closely together and maintaining a separation that minimizes the interference effects.[10] This provides a compact optical setup, comprising only the diffractive optical element (DOE) and a Fourier transforming lens, as shown in Fig. 6.

3.1 *Designing the Output*

The number of ''on'' diffraction orders in the final design is set by the maximum intensity allowable $I_{\max}$ and the intensity produced by the laser source I_{in}:

$$N_T=\eta\frac{I_{\text{in}}}{I_{\max}}, \tag{14}$$

where N_T is the number of ''on'' diffraction orders, and η is the estimated diffraction efficiency of the element. The number of orders required across the diameter of each fiber core is given by

$$N_D=\sqrt{\frac{N_T}{\pi N_f}}, \tag{15}$$

where N_f is the number of fibers being used.

The separation is then calculated taking into account the core diameter of the fiber and the necessity of maintaining the relative separation σ_R of the ratio of separation to spot size, at

$$\sigma_R=(j+0.229), \tag{16}$$

where j is any integer greater than one.[10]

The separation can then be used to determine how many ''off'' diffraction orders are required between each fiber. This output is then used as the target intensities in an optimization algorithm to find the desired phase profile. In the following examples, we used the modified GS algorithm presented earlier to optimize the phase profiles.

3.2 *Examples of Coupling Elements*

The design brief for the following examples was to produce diffractive elements for coupling to 2, 4, 8, and 10 individually mounted fibers with 400-μm core diameters. The operating wavelength of the 300-mJ laser used in conjunction with these elements was 1064 nm and the minimum feature size allowed for the designs was 2 μm. The connector specifications are shown in Fig. 7. The 8- and 10-way elements used the same connector with the 8-way fan-out utilizing only the fibers mounted in a ring around the edge of the connector. The 2- and 4-way fan-outs had to be able

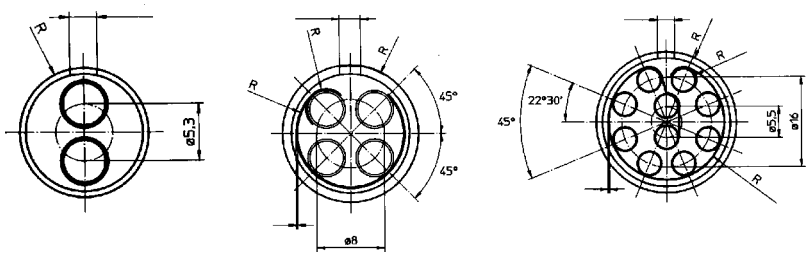


Fig. 7 Design specification for the connectors used with the fan-out elements.

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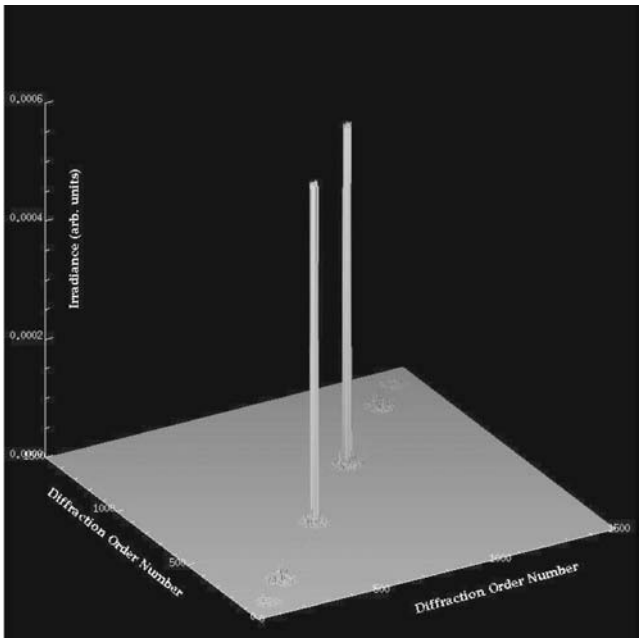


Fig. 8 Simulated output from the two-way fan-out element. Each set of diffraction orders contains 33 orders across the diameter.

to couple 100 mJ of light with a pulse length of 20 ns into the fibers while producing a peak intensity of no more that 4.5 GW/cm^2. The 8- and 10-way fan-outs had to couple a 300-mJ pulse of the same duration while satisfying the intensity constraint.

3.2.1 *Two-way fan-out*

To reduce the individual intensity of the diffraction orders below 4.5 GW/cm^2 we illuminated the input face of each fiber with flat-top illumination containing 33 diffraction orders in diameter. The two sets of on orders were encoded in a signal window containing 375×375 diffraction orders. The diffractive element was designed as a 16-phase-level element containing 1500×1500 pixels. The simulated element performance has a diffraction efficiency of 90.35% with a uniformity error of 3.50%. The simulated output from the element is shown in Fig. 8.

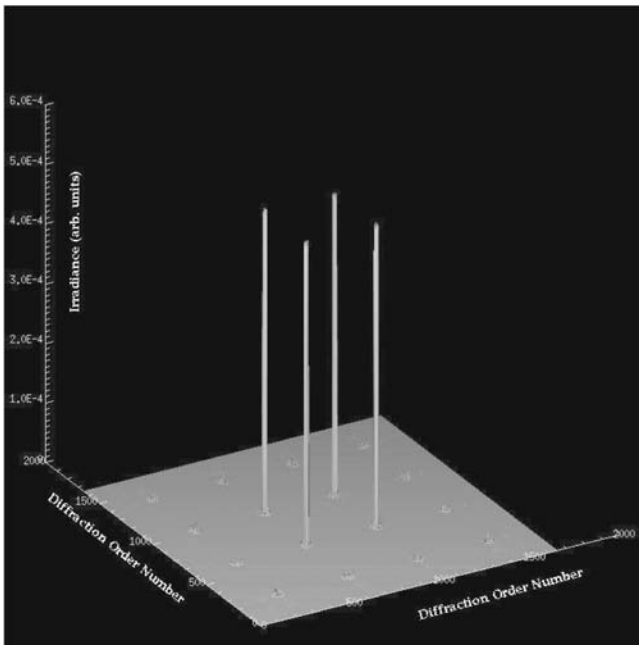


Fig. 9 Simulated output from the four-way fan-out element. Each set of diffraction orders contains 25 orders across the diameter.

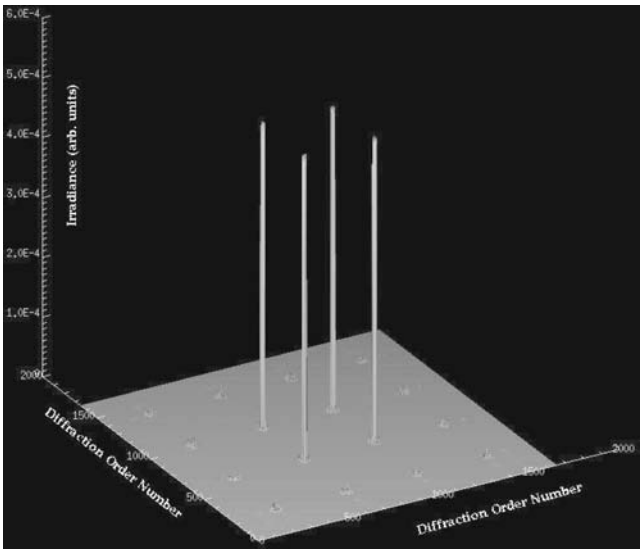


Fig. 10 Simulated output from the eight-way fan-out element. Each set of diffraction orders contains 29 orders across the diameter.

3.2.2 *Four-way fan-out*

Due to the increase in the number of fibers used to transmit the light, the number of orders required to illuminate each fiber face was reduced to 25 orders in diameter. This reduction is desirable as it enables an increased spacing between the diffraction orders, minimizing the number that must be considered in the design. The signal window contained 420×420 diffraction orders and the phase profile was encoded into 16 phase levels on a grid containing 1650

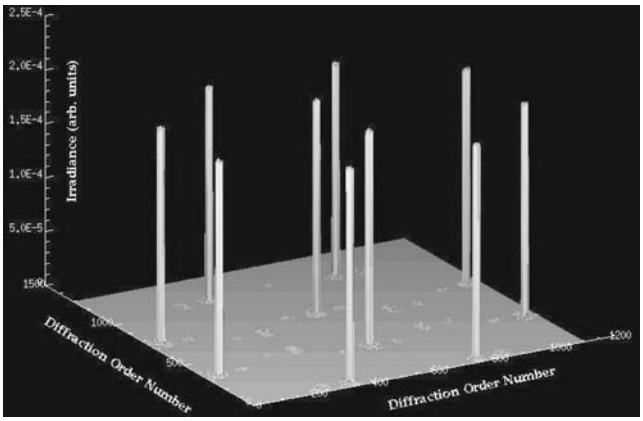


Fig. 11 Simulated output from the 10-way fan-out element. Each set of diffraction orders contains 25 orders across the diameter.

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$\times 1650$ pixels. The final design has an efficiency of 88.51% with a uniformity error of 2.8% with the simulated performance shown in Fig. 9.

3.2.3 *Eight-way fan out*

Applying the intensity constraint to the design of the eight-way fan out led to the required illumination containing 29 diffraction orders in diameter. The size of the connector determined a minimum signal window size of 1200×1200 diffraction orders. Using an element with 16 phase levels would have required a phase profile of nearly 5000×5000 pixels. While the design of this element was possible, fabrication difficulties prevented the implementation. As a result, the number of phase levels was increased to 32, trading fabrication complexity for a reduction in the number of pixels. The resulting increase in the minimum feature size improves the alignment tolerance during the fabrication process. The final element design was a 32 phase level element containing 2400×2400 pixels. The simulated efficiency of the element is 87.22% and the uniformity error is only 2.51%. The output from the element is shown in Fig. 10.

3.2.4 *Ten-way fan-out*

The 10-way fan-out element was designed with the same considerations as the 8-way element with a reduction in the number of diffraction orders corresponding to the increase in the number of phase levels. The sets of diffraction orders, containing 25 orders in diameter, were encoded in a signal window of 1040×1040 diffraction orders and a 32-phase-level element was produced containing 2048×2048 pixels. The simulated performance, shown in Fig. 11, displays a diffraction efficiency of 86.99% with a uniformity error of 2.59%.

4 Further Optimization and Integration of an Element and a Lens

The preceding design results can be improved further by optimizing the initial phase estimations for the GS algorithm. Our best design results are that, for square and circular SG beam shaping, the MSEs of the distributions fitted to the corresponding 12th-order SG beams are 1.26×10^{-3} and 6.54×10^{-4}, and the bucket energy contained within the desired region is 98.69 and 99.45%. The detail methods will be reported elsewhere. Furthermore, to avoid the degraded performance caused by misalignment between the DOE and a lens, we add a quadratic lens's phase to the far-field beam-shaping element. Figure 12 shows the integrated beam-shaping elements for square and circular output beams. Figure 13 shows the simulated square and circular output beams. The MSEs of the two distributions produced by the integrated elements are 4.0×10^{-3} and 2.3×10^{-3}, while the bucket energies are 99.23 and 99.53%. The reasons for the changes of the figures might be the decrease in the sample points of the simulated output beams and the error introduced by the quadratic lens function.

The integrated beam-shaping elements are tolerant to the aberrations of the incident beam, the phase quantization of the element and the etching errors. Figure 14(a) shows the simulated circular output beams of a 16-phase-level quantized element. The MSE and the bucket energy are 2.1×10^{-3} and 98.21%. When the etching error of the element is $\pm 10\%$, the simulated beams are shown in Figs. 14(b) and 14(c). The MSEs of the two distributions are 3.7×10^{-3} and 1.6×10^{-3}, while the bucket energies are 94.88 and 95.32%, respectively. Furthermore, the adverse effects caused by misalignment of the element or any lateral beam-element displacement can easily be eliminated by monitoring the output beam intensity distribution.

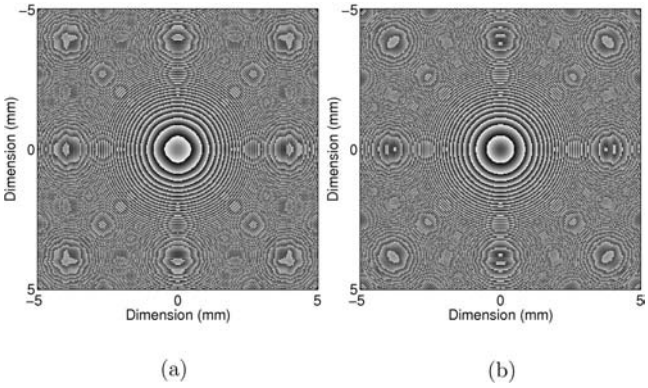


Fig. 12 Integrated beam-shaping elements for (a) a square output beam and (b) a circular output beam, where the diffractive distance is 340 mm. The range of variation of the phase is from $-\pi$(white) to π(black). The pattern is consistent with the final mask used for exposure, i.e., large phase values correspond to thicker element profiles. The simulated diffractive output beam is shown in Fig. 13.

5 Conclusions

In summary, we presented a modified iterative Fourier transform algorithm for designing DOEs for far-field free-space laser beam shaping. The algorithm begins with the GS approach to obtain a stable solution followed by several new iterations. The main features of the new iterations are first, the phase distribution at every new iteration begins with the results of GS and remains unchanged, and second, the modified constraint function is imposed on the Fourier domain of the iterations. The modified Fourier-domain constraint function is based on the difference between the output results of GS or the output results of new iterations, and the required ideal output profile. Generally, only a few new iterations can produce an excellent solution that is indicated by high efficiency and low-error output. For SG beam shaping, the MSE of the intensity distribution fitted to a 12th-

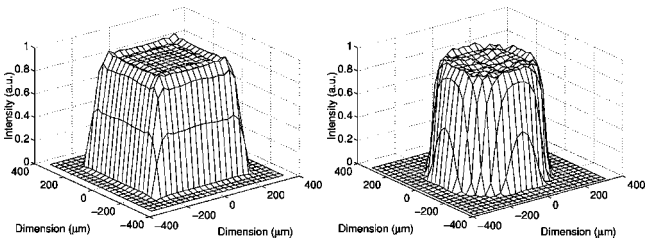


Fig. 13 Simulated square and circular output beams produced by the elements shown in Fig. 12. The MSEs of the two distributions are 4.0×10^{-3} and 2.3×10^{-3}, while the bucket energies are 99.23 and 99.53%.

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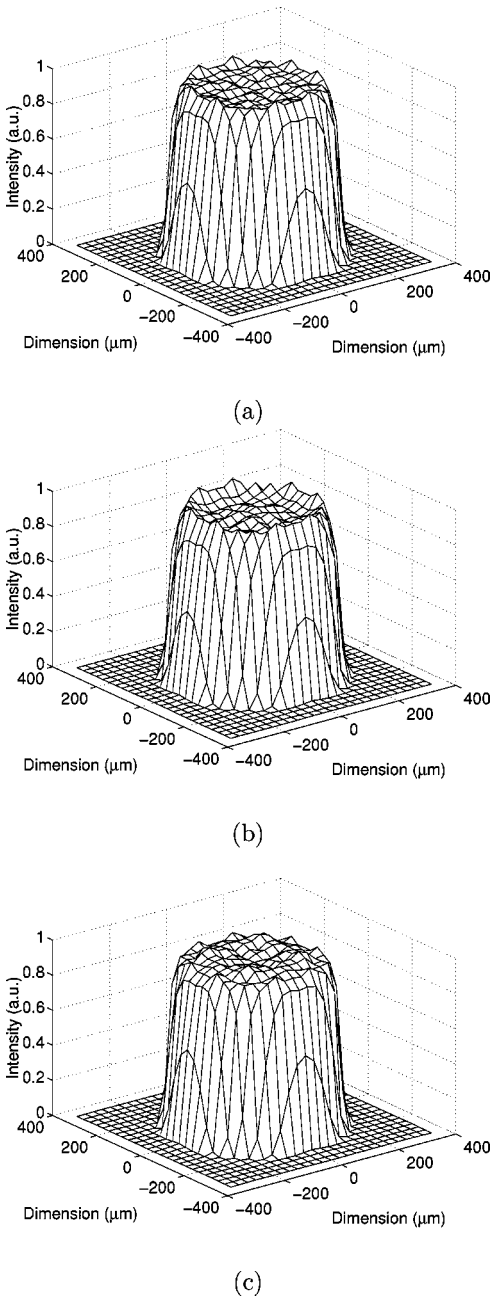


Fig. 14 Simulated output beams produced by (a) a 16-phase-level quantized element, (b) a quantized element with 10% etching error, and (c) a quantized element with −10% etching error. The MSEs of the three distributions are 2.1×10^{-3}, 3.7×10^{-3}, and 1.6×10^{-3}. The bucket energies are 98.21, 94.88, and 95.32%, respectively.

order SG beam is 9.75×10^{-3} with 97.4% of the incident energy containing in the desired region. Free-space beam-shaping elements are also presented combining beam conditioning and fan-out functions in a single element for coupling into multimode optical fibers. These elements, which can contain over 2000×2000 pixels, display diffraction efficiencies of over 85% while maintaining uniformity errors of $<5\%$.

To avoid the degradation in performance caused by misalignment between the DOE and a lens for the SG beam-shaping case, we can add the phase of a quadratic lens to the far-field beam-shaping element. The integrated beam-shaping elements are tolerant to phase quantization and etching errors. The MSE even slightly improves (decreases) when the elements are quantized to 16 phase levels and when a negative etch depth error is introduced, while the bucket energy slight decreases.

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